

Ingestion Architecture & Streaming Pipelines

Target Platform: Google Cloud
Platform (GCP)

System of Record:
ServiceNow ITSM

Version: 2.0 (Production
Blueprint)

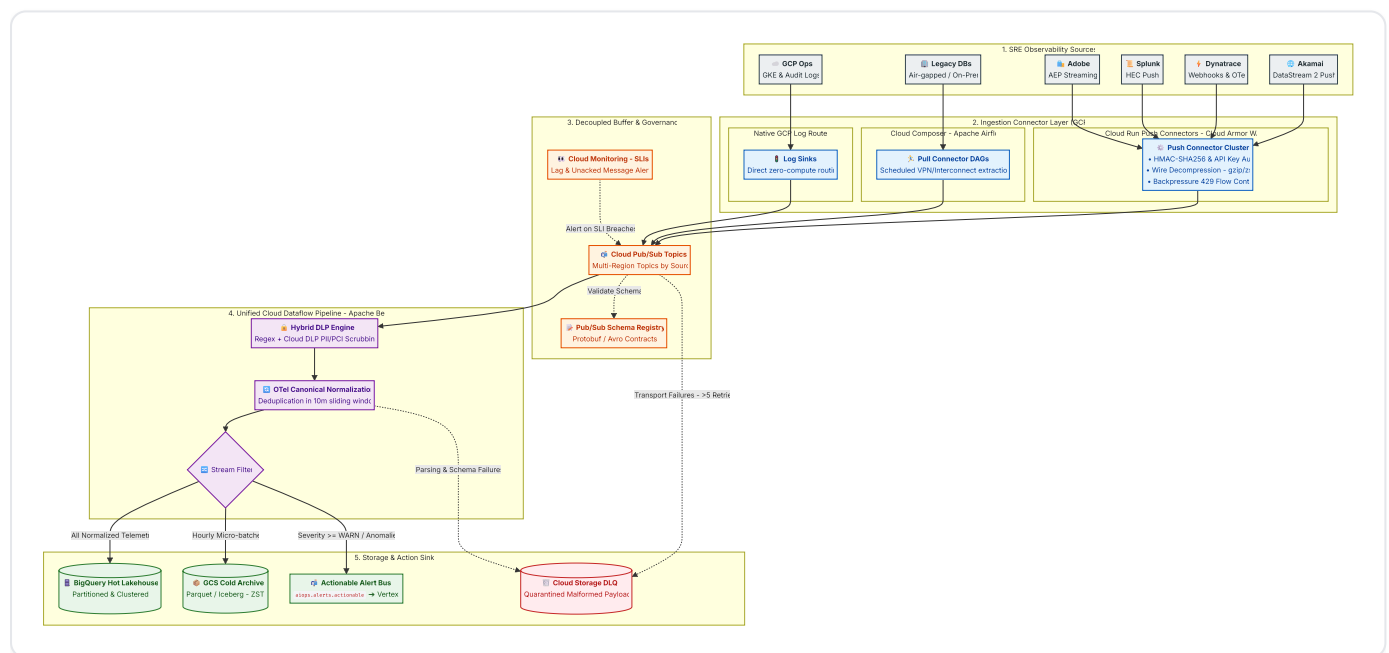
Generated:
August 2026

Ingestion Layer & Streaming Pipelines Architecture

This document outlines the **Enterprise AIOPS Ingestion Engine**, which captures high-throughput telemetry from multi-cloud and edge observability sources, standardizes and sanitizes it in real time, and routes it to downstream analytics and incident response systems natively on **Google Cloud Platform (GCP)**.

1. High-Level Ingestion Architecture

The ingestion architecture follows a decoupled, connector-based pattern capable of handling 150,000 to 500,000+ Events Per Second (EPS) with sub-second latency and zero data loss.



2. The Ingestion Connector Layer

Telemetry sources exhibit different networking capabilities and security constraints. We utilize three distinct connector patterns to ingest telemetry securely into GCP:

2.1 Push Connectors (Modern SaaS & Edge Ingress)

Used by sources capable of real-time outbound streaming (**Akamai DataStream 2**, **Dynatrace Webhooks**, **Splunk HEC**, **Adobe Analytics AEP**).

- **Ingress Security:** Traffic passes through **Cloud Armor WAF** with strict IP allowlisting (filtering for vendor CIDR ranges) and volumetric DDoS rate limiting.
- **Compute Layer:** Auto-scaling, stateless **Cloud Run** container services deployed across multiple zones.
- **Authentication & Webhook Validation:**
 - **API Keys & Tokens:** Verified dynamically against **GCP Secret Manager** with in-memory caching.
 - **HMAC-SHA256 Signatures:** Validates webhook payload authenticity using vendor signature headers (e.g., `X-Akamai-Signature` , `X-Adobe-Signature`) to prevent request spoofing.
- **Wire Compression:** Enforces `Content-Encoding: gzip` / `zstd` on inbound streams, reducing cross-cloud data transfer volume and egress bandwidth costs by up to 70%.
- **Backpressure & Flow Control:** If downstream Pub/Sub publishing encounters transient backpressure, Cloud Run returns `HTTP 429 (Too Many Requests)` or `HTTP 503` with a `Retry-After` header, prompting upstream SaaS forwarders to buffer and retry exponentially.

2.2 Pull Connectors (Legacy & Air-Gapped Systems)

Used by on-premise relational databases, mainframes, or network appliances that cannot push outbound traffic.

- **Orchestration:** **Cloud Composer (Managed Apache Airflow)** schedules and executes extraction DAGs.
- **Networking:** Traffic travels over private, encrypted tunnels via **Cloud VPN** or **Cloud Interconnect**.
- **Extraction & Retries:** Airflow operators extract delta records, convert them to canonical JSON/Avro, and publish to Cloud Pub/Sub with automatic exponential backoff retries.

2.3 Native Connectors (GCP Operations Suite)

Used natively by Google Cloud Platform workloads (**Google Kubernetes Engine (GKE)**, **Cloud Run**, **Cloud Audit Logs**).

- **Mechanism:** Leverages **Cloud Logging Log Router Sinks** pointing directly to Cloud Pub/Sub topics.
 - **Advantage:** Zero-compute, zero-maintenance ingestion with sub-second delivery latency and zero egress cost.
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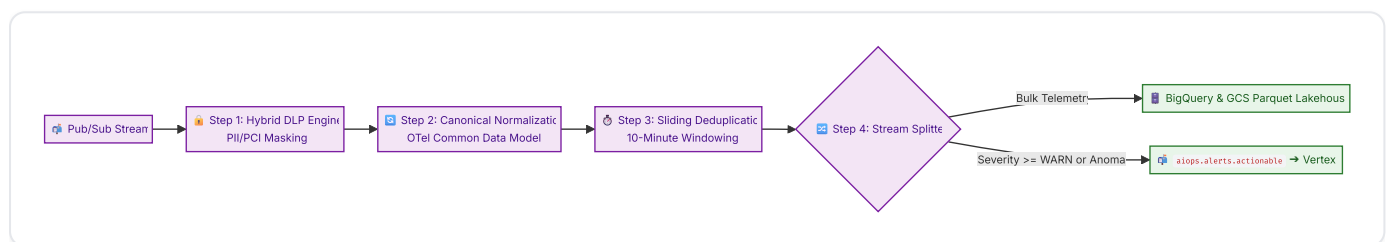
3. Decoupled Buffer & Stream Processing

3.1 Cloud Pub/Sub (The Multi-Region Event Bus)

- **Topic Isolation:** Dedicated, isolated topics per source (e.g., `telemetry.akamai.raw`, `telemetry.dynatrace.raw`, `telemetry.splunk.raw`) prevent single-source traffic storms (such as a DDoS attack logged by Akamai) from starving other telemetry streams.
- **Multi-Region Durability:** Topics are configured with multi-region message routing policies to ensure continuous ingestion availability during single-region GCP outages.

3.2 Cloud Dataflow (Unified Apache Beam Streaming Pipeline)

A unified, horizontally auto-scaling streaming pipeline processes incoming telemetry in four sequential stages:



1. **Hybrid DLP Engine:** Fast, in-memory regex tokenizers redact high-risk payment card numbers (PCI-DSS) and social security numbers. Unstructured log bodies are asynchronously scrubbed using the **Cloud Data Loss Prevention (DLP) API**.
2. **Canonical Normalization:** Standardizes disparate formats (Akamai JSON, Splunk HEC, Dynatrace PurePath, Adobe clickstream) into an **OpenTelemetry-aligned canonical schema** (`aiops_lakehouse.telemetry_canonical`).
3. **Sliding Window Deduplication:** Removes duplicate events within a 10-minute sliding window based on (`source_tool`, `source_event_id`, `timestamp`).
4. **Intelligent Stream Splitting:**
 - **Bulk Analytical Stream:** Routes 100% of clean telemetry to BigQuery (Hot Tier) and GCS Parquet (Cold Tier).
 - **Actionable Alert Stream:** High-severity anomalies and warning events (`severity >= WARN` or Davis AI problems) are extracted and published to the high-priority Pub/Sub topic `aiops.alerts.actionable` for consumption by the **Vertex AI Semantic Router**.

4. Reliability, Robustness & Governance Patterns

4.1 Schema Management & Dynamic Evolution

- **Pub/Sub Schema Registry:** Enforces Protobuf and Avro schema contracts at topic ingress.
- **Schema Drift Protection:** Incoming records that introduce backward-compatible fields are parsed dynamically into `raw_attributes JSON`, preventing Dataflow worker crashes while preserving full fidelity.

4.2 Multi-Tier Dead-Letter Queues (DLQs)

To prevent unparseable or poisoned payloads from blocking the pipeline:

1. **Transport-Level DLQ (Pub/Sub):** If Dataflow fails to acknowledge a message after 5 delivery attempts, Pub/Sub diverts it to a native dead-letter topic (`telemetry.*.dlq`).
2. **Application-Level DLQ (Dataflow Side-Output):** Payloads that parse structurally but fail semantic validation (e.g., corrupted timestamps) are emitted via Beam side-outputs.
3. **Storage & Replay:** Both DLQ streams persist to a date-partitioned **Cloud Storage DLQ Bucket** (`gs://aiops-dlq-bucket/YYYY/MM/DD/`) for offline inspection, alerting, and automated replay.

4.3 Ingestion Service Level Indicators (SLIs) & Alerting

The platform team continuously monitors pipeline health in **Cloud Monitoring**:

Metric Name	Target SLI	Alert Severity	Action upon Breach
<code>oldest_unacked_message_age</code>	\$< 5\$ minutes	P1 (Critical)	Pages Data Platform SRE via ServiceNow webhook
<code>dataflow/system_lag</code>	\$< 120\$ seconds	P2 (Major)	Triggers automated worker auto-scaling
<code>dataflow/watermark_age</code>	\$< 180\$ seconds	P2 (Major)	Alerts on upstream source timestamp drift
<code>dlq_message_count</code>	\$< 100\$ msgs/hour	P3 (Warning)	Flags upstream schema violation for review

5. Related Architecture Specifications

- [SRE Observability Fleet - Source Telemetry Matrix](#)
- [Data Contracts & Canonical Schemas](#)
- [Ingestion Best Practices Guide](#)
- [Unified Lakehouse & AI Feature Store](#)